

Multimodal Natural Language Processing

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- ▶ Traditional NLP models are **unimodal**: text in, text out; they can neither see nor hear the world their language describes.
- ▶ Human communication is grounded in perception: much of what language refers to arrives through vision, sound, and the other senses.
- ▶ **Vision-language models** align images with text at web scale, and today's assistants are **multimodal LLMs** built on top of that alignment.
- ▶ This lecture builds from contrastive alignment (CLIP and its successors) to the projector-and-instruction recipe behind multimodal assistants, and on to the natively multimodal and omni models of 2025–2026.

multimodal adjective

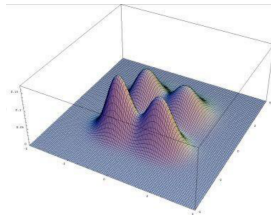
[muhl-tee-moh-duhl]

: having or involving several modes, modalities, or maxima

multimodal distributions

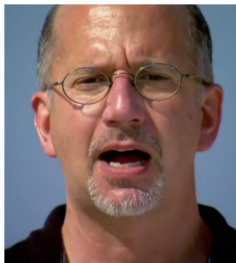
multimodal therapy

In our case, focusing on NLP: text + one or more other *modality* (images, speech, audio, olfaction, others). We'll mostly focus on images as the other modality.



A multimodal distribution: several maxima.
Definition: Merriam-Webster,
"multimodal".

- ▶ **McGurk effect** (McGurk & MacDonald, 1976)
 - Classic demonstration of interaction between modalities (audio + vision) in perception
 - Speech sound is misheard depending on visual (lip movement) cues
- ▶ [Watch this demonstration on YouTube](#)



Source: frame from “McGurk effect: Auditory Illusion”, BBC *Horizon*.

- ▶ **The symbol grounding problem** (Harnad, 1990): “How can the semantic interpretation of a formal symbol system be made *intrinsic* to the system, rather than just parasitic on the meanings in our heads?” Harnad likens learning from symbols alone to learning Chinese from a Chinese-Chinese dictionary.
- ▶ **The octopus test** (Bender & Koller, ACL 2020): a deep-sea octopus O taps the telegraph cable between two islanders and learns to impersonate one of them purely from the statistics of their messages. The act holds until real-world help is needed: “having only form available as training data, O did not learn meaning.”
- ▶ Their conclusion: a system trained only on *form* has a priori no way to learn *meaning*; to learn language, it “must solve what Harnad (1990) calls the symbol grounding problem.”
- ▶ Multimodal NLP is the constructive response: tie language to what the system can see and hear.

Harnad, *Physica D* 42:335–346, 1990; Bender & Koller, ACL 2020, aclanthology.org/2020.acl-main.463.

- 2015** Captioning and VQA become benchmark tasks: Show and Tell (CVPR 2015), VQA (ICCV 2015)
- 2019** BERT-style vision-language pretraining: ViLBERT (NeurIPS 2019), LXMERT (EMNLP 2019)
- 2021** Contrastive alignment at web scale: CLIP (ICML 2021)
- 2022** Few-shot multimodal in-context learning: Flamingo (NeurIPS 2022)
- 2023** Visual instruction tuning (LLaVA); frontier chatbots gain vision (GPT-4V system card, Sep 2023)
- 2024** Natively multimodal frontier models: GPT-4o, one network end-to-end across text, vision and audio
- 2025** Images enter the chain of thought (o3); open omni models; native image generation
- 2026** Native multimodality becomes the default in frontier and open releases alike

Every milestone above is cited in the references.

- ▶ Explain why language needs grounding, and map the vision-language task space to its canonical benchmarks.
- ▶ Explain contrastive vision-language alignment (CLIP, SigLIP) and what dual encoders can and cannot do.
- ▶ Describe visual instruction tuning, the projector recipe behind multimodal assistants, and how images become tokens.
- ▶ Survey the 2025–2026 generation: open-weight flagships, natively multimodal frontier systems, unified understanding-and-generation models, and omni models.
- ▶ Explain the applications beyond chat: video, documents, multimodal retrieval, GUI agents, and vision-language-action robotics.
- ▶ Interpret multimodal benchmarks (MMMU, MMMU-Pro) and identify standing failure modes such as object hallucination, visual token cost, and multimodal jailbreaks.
- ▶ Compare early, middle and late fusion, and the classic challenges of multimodal integration.

Vision-Language Models

Parts of this section adapted from D. Kiela, "Multimodal Deep Learning (with an NLP focus)",
guest lecture, Stanford CS224n, Winter 2023.

- **Visual question answering:** free-form questions about an image (VQA, ICCV 2015). VQAv2 (CVPR 2017) pairs every question with two images that yield two different answers, so language priors alone cannot score well.
- **Captioning:** describe the image (COCO Captions: five human captions per image, CIDEr and BLEU scoring).
- **Referring expressions:** localize “the man in the red hat” (RefCOCO family, 2016).
- **Reading text in images:** TextVQA (CVPR 2019).
- **Documents and charts:** DocVQA (WACV 2021), ChartQA (ACL Findings 2022).



VQAv2's complementary pairs: same question, two images, two answers. Source: Goyal et al., “Making the V in VQA Matter”, CVPR 2017, [arXiv:1612.00837](https://arxiv.org/abs/1612.00837), Figure 1.

Multimodal LLMs are still evaluated on descendants of all five tasks.

- **Contrastive Language-Image Pretraining** (Radford et al., ICML 2021): a dual encoder trained on 400M web (image, text) pairs so that each image ranks its own caption above the other captions in the batch, and each caption ranks its own image.
- **What it buys:** zero-shot classification from text prompts, image-text retrieval, and robustness to sketches, renditions and adversarial photographs that break supervised ImageNet models.
- **What it cannot do:** it only *scores* image-text pairs. There is no decoder, so it cannot describe an image, answer a question, or hold a conversation.
- **Why it still matters:** contrastively pretrained encoders became the eyes of nearly every multimodal LLM that followed.

Radford et al., “Learning Transferable Visual Models From Natural Language Supervision”, ICML 2021,
[arXiv:2103.00020](https://arxiv.org/abs/2103.00020).

For a batch $\mathcal{B}$ of ℓ_2 -normalized image embeddings x_i and text embeddings y_i , CLIP minimizes a symmetric cross-entropy over temperature-scaled cosine similarities:

$$\mathcal{L}_{\text{softmax}} = -\frac{1}{2|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \left(\log \frac{e^{t x_i \cdot y_i}}{\sum_j e^{t x_i \cdot y_j}} + \log \frac{e^{t x_i \cdot y_i}}{\sum_j e^{t x_j \cdot y_i}} \right)$$

The temperature t is *learned*, and both terms need the full batch similarity matrix. SigLIP replaces this with an independent binary decision per pair:

$$\mathcal{L}_{\text{sigmoid}} = -\frac{1}{|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \sum_{j=1}^{|\mathcal{B}|} \log \frac{1}{1 + e^{z_{ij} (-t x_i \cdot y_j + b)}}, \quad z_{ij} = \begin{cases} 1 & i = j \\ -1 & i \neq j \end{cases}$$

with a learned bias b offsetting the imbalance of negative pairs. Sigmoid training wins clearly below batch size 16k; both losses saturate around 32k.

Notation follows Zhai et al., ICCV 2023, [arXiv:2303.15343](https://arxiv.org/abs/2303.15343), Secs. 3.1–3.2 (their formalization of the CLIP loss).

- **SigLIP** (Zhai et al., ICCV 2023) replaces CLIP's softmax contrastive loss with a pairwise **sigmoid** loss: every image-text pair becomes an independent binary decision, so the loss needs no global view of all pairings in the batch.
- **SigLIP 2** (Feb 2025) folds captioning-based pretraining, self-distillation, masked prediction and online data curation into one multilingual recipe, with much stronger localization and dense features.
- Its **NaFlex** variants support several resolutions and preserve the input's native aspect ratio; checkpoints range from ViT-B (86M) to g (1B).
- These encoders, more than CLIP itself, are what current open multimodal models plug in: LLaVA-OneVision and Janus-Pro use SigLIP, BAGEL uses SigLIP 2.

Zhai et al., [arXiv:2303.15343](#); Tschannen et al., [arXiv:2502.14786](#).

Before decoder LLMs, vision-language pretraining meant BERT-style encoders over text and image features:

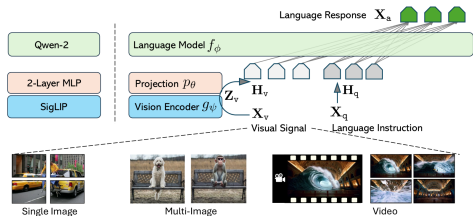
- **VisualBERT** (2019): one Transformer stack attends over text tokens and object-detector regions together (early fusion); masked language modelling *with the image*.
- **ViLBERT** (NeurIPS 2019) and **LXMERT** (EMNLP 2019): two streams, one per modality, exchanging information through co-attention layers.
- **Pixel-BERT** (2020): drops the object detector; a CNN feeds pixel-level features straight to the Transformer.
- **FLAVA** (CVPR 2022): three towers (vision, language, multimodal) trained with masked and contrastive objectives on public data; one model for unimodal and multimodal tasks.
- **Why they faded:** they classify, score and retrieve over fixed label sets, each downstream task needs its own fine-tuned head, and none of them generates language.

- **Nomenclature.** An **LLM** is language-only. A **VLM** (vision-language model) adds a vision encoder so the model can condition on images. An **MLLM** (multimodal LLM) spans several modalities, such as image, video and audio, in one model.
- **What the models above can and cannot do.** CLIP scores an image against candidate texts; the Visual BERTs classify or answer from a fixed label set. None of them writes an answer. A model that has to *describe* an unfamiliar chart, or hold a conversation about a photograph, needs a language decoder.
- **The shift.** Attach a pretrained vision encoder to a pretrained decoder LLM through a small trained module, the **projector** (also called the connector), and train mostly that module. Flamingo (2022), BLIP-2 (2023) and LLaVA (2023) established the recipe; Qwen3-VL and InternVL3.5 are its open-weight production descendants.
- Frontier systems have since pushed further, toward native multimodality (later in this lecture). The connector variants are compared in detail later in the course.

- **The missing ingredient was data.** Instruction tuning made text LLMs conversational, but there was no multimodal equivalent: web image-text pairs are captions, not conversations.
- **LLaVA's trick (Liu et al., 2023):** feed image *captions and bounding boxes* to text-only GPT-4 and let it synthesize 158k instruction examples (conversations, detailed descriptions, and complex-reasoning QA) about images it never saw.
- **Two-stage recipe:** first train only the projector, so visual tokens land in the LLM's embedding space (feature alignment); then fine-tune projector and LLM together on the instruction data. The vision encoder stays frozen throughout.
- The result is a model that can hold a conversation about a photograph, and the template (synthetic data included) behind the open multimodal assistants that followed.

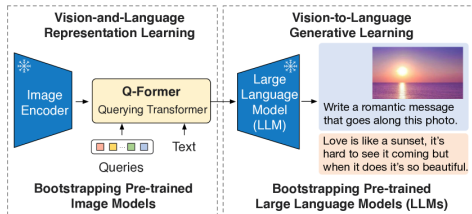
Liu et al., "Visual Instruction Tuning", NeurIPS 2023, [arXiv:2304.08485](https://arxiv.org/abs/2304.08485).

- **LLaVA-1.5** (CVPR 2024): a two-layer MLP projector plus academic VQA data reaches state of the art on 11 benchmarks with $\sim 1.2\text{M}$ public samples and about a day of training on one 8-GPU node.
- **LLaVA-NeXT** (2024): tiles the input into 336px crops (“AnyRes”) for up to $4\times$ more pixels, plus OCR and chart data.
- **LLaVA-OneVision** (2024): one open model for single images, multi-image input and video.
- **Molmo and PixMo** (CVPR 2025): open weights *and open data*, collected without distilling any proprietary VLM; the 72B model beat Claude 3.5 Sonnet and Gemini 1.5 Pro.



The recipe in 2024: SigLIP encoder, MLP projector, Qwen-2 LLM. Source: Li et al., “LLaVA-OneVision”, [arXiv:2408.03326](https://arxiv.org/abs/2408.03326), Figure 1.

- **Patches are the atoms:** a ViT cuts the image into fixed-size patches. ViT-L/14 at 336×336 yields $(336/14)^2 = 576$ patch embeddings, the per-image token count of LLaVA-1.5.
- **Compression is a design axis:** BLIP-2's Q-Former distills every image into 32 learned query tokens; Flamingo's Perceiver Resampler emits 64.
- **Resolution multiplies cost:** AnyRes encodes each 336px tile separately; Qwen's native dynamic resolution merges each 2×2 patch group and spends tokens in proportion to image area.
- The trade-off between visual fidelity and context budget shapes every design in this lecture.



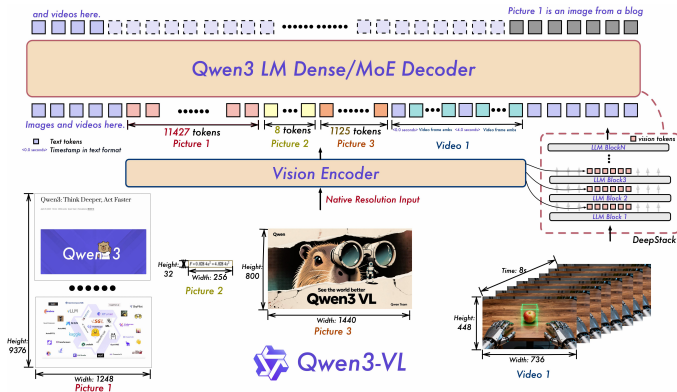
BLIP-2's Q-Former: 32 learned queries distill a frozen encoder's features for a frozen LLM. Source: Li et al., BLIP-2, ICML 2023, [arXiv:2301.12597](#), Figure 1.

Multimodal LLMs in 2025–2026

- **Qwen2.5-VL** (Feb 2025): a dynamic-resolution ViT trained from scratch with window attention; absolute time encoding gives hours-long video comprehension with second-level event localization; grounding by bounding box or point; structured parsing of invoices, forms and tables.
- **Qwen3-VL** (Sep–Nov 2025): dense and MoE sizes from 2B to 235B, each in Instruct and reasoning-focused Thinking editions; native 256K context, expandable to 1M; operates PC and mobile GUIs as a visual agent; OCR in 32 languages.
- **InternVL3 / InternVL3.5** (2025): *native multimodal pre-training*, one joint stage over text and multimodal corpora instead of retrofitting a finished text LLM; InternVL3.5 adds Cascade RL and a Visual Resolution Router for a $4\times$ inference speedup, reporting state of the art among open-source MLLMs.

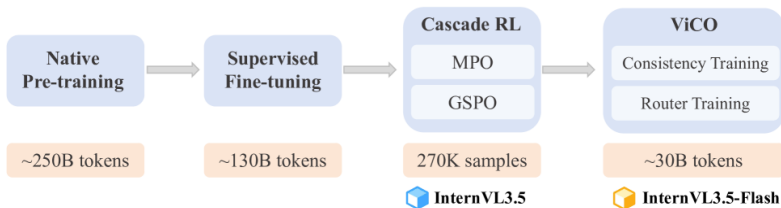
Bai et al., [arXiv:2502.13923](#); Bai et al., [arXiv:2511.21631](#); Zhu et al., [arXiv:2504.10479](#); Wang et al., [arXiv:2508.18265](#).

Inside a 2025 Flagship: Qwen3-VL



Native-resolution images and timestamped video frames enter one decoder as interleaved tokens, so token count scales with input size: 11,427 tokens for a long webpage, 8 for a small formula. DeepStack feeds multi-level ViT features into several LLM blocks. Source: Qwen team, Qwen3-VL repository, [arXiv:2511.21631](https://arxiv.org/abs/2511.21631).

- **Encoder first:** Qwen2.5-VL trains its ViT alone on 1.5T tokens of captions, OCR and visual knowledge while the LLM stays frozen, then trains ViT and LLM jointly on 2T mixed multimodal tokens.
- **Then supervised fine-tuning** on multimodal instruction data, exactly as for text LLMs.
- **Then reinforcement learning:** InternVL3.5's Cascade RL runs an offline preference stage (MPO) and then an online stage (GSPO). RL post-training algorithms are covered later in the course.



InternVL3.5's recipe: native pre-training, supervised fine-tuning, Cascade RL; the final ViCO stage trains the Flash variant's resolution router. Source: Wang et al., [arXiv:2508.18265](https://arxiv.org/abs/2508.18265), Figure 3.

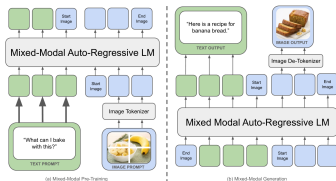
- **GPT-4o** (May 2024): “a single new model end-to-end across text, vision, and audio”; the same network reads and produces all of them.
- **Gemini 2.5 Pro** (Mar 2025): text, audio, images, video and whole code repositories in a 1M-token context window.
- **GPT-5** (Aug 2025): one unified system that decides when to think longer; 84.2% on MMMU at launch.
- **Gemini 3** (Nov 2025): launched as “the best model in the world for multimodal understanding”: 81% on MMMU-Pro, 87.6% on Video-MMMU.
- **2026 so far**: new releases land every few months (GPT-5.4 through 5.6, Gemini 3.5 and 3.6, Claude Opus 4.7 and Fable 5); the best frontier MMMU-Pro score moved from 81% to 83% without tools (84.6% with tools) in eight months.

Sources: OpenAI, Google and Anthropic release announcements, 2024–2026 (see references).

- **OpenAI o3 and o4-mini** (Apr 2025) were the first OpenAI models that “think with images in their chain-of-thought” rather than merely looking at them.
- During reasoning the model crops, zooms into and rotates the user’s image, natively and without separate specialist models, and can mix these steps with web search and Python.
- Result: state of the art at the time on MMMU, MathVista and chart reasoning, and 95.7% on the V* visual-search benchmark, “largely solving” it.
- The pattern generalizes: perception becomes a sequence of *actions* on the input, interleaved with textual reasoning. How such reasoning is trained is covered later in the course.

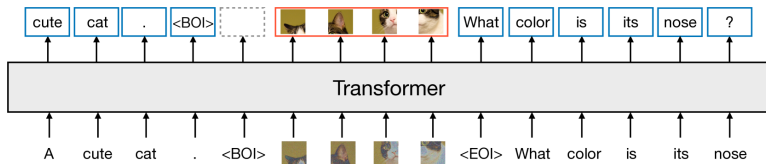
OpenAI, “Thinking with images”, April 2025, openai.com/index/thinking-with-images.

- The projector recipe bolts vision onto a finished LLM. The native alternative pretrains one transformer on every modality tokenized together: **Chameleon** (Meta, 2024) maps a 512×512 image to 1024 discrete tokens, and one autoregressive model reads and writes both modalities.
- **Llama 4** (Apr 2025): Meta's first open-weight family pretrained with early fusion, jointly on text, image and video data.
- Across 457 trained models, late fusion showed no inherent advantage; early fusion is stronger at small scales and cheaper to train (Shukor et al., ICCV 2025).



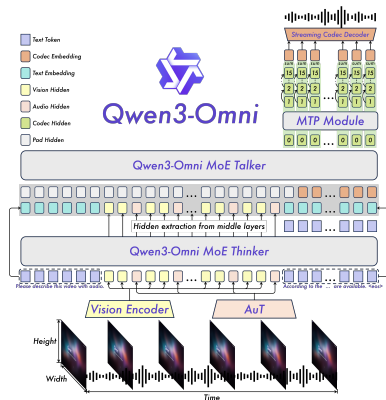
Source: Chameleon Team, [arXiv:2405.09818](https://arxiv.org/abs/2405.09818), Figure 1.

- **Transfusion** (ICLR 2025): one transformer, two losses: next-token prediction on text tokens, a diffusion objective on continuous image patches in the same sequence; scales better than quantizing images.
- **Janus-Pro** (DeepSeek, Jan 2025): decoupled visual encoding: SigLIP features for understanding, VQ tokens for generation, one shared autoregressive core.
- **In production:** GPT-4o image generation (Mar 2025) and Gemini 2.5 Flash Image (“Nano Banana”, Aug 2025) put native image generation and editing in assistants; Nano Banana Pro (Nov 2025) grounds generations in live search.



One sequence mixes text tokens with continuous image patches between <BOI>/<EOI>. Source: Zhou et al., ICLR 2025, [arXiv:2408.11039](https://arxiv.org/abs/2408.11039), Figure 1.

- **GPT-4o** answers speech in 320 ms on average; the cascaded pipeline it replaced (transcription, text LLM, TTS) took 2.8–5.4 s and lost tone, multiple speakers and laughter.
- **Qwen2.5-Omni** (Mar 2025): **Thinker-Talker**: the Thinker LLM writes text while the Talker streams speech tokens from its hidden states; TMRoPE keeps audio and video frames on one timeline.
- **Qwen3-Omni** (Sep 2025): both parts become MoE; 234 ms theoretical first-packet latency; open-source state of the art on 32 of 36 audio and audio-visual benchmarks. **Qwen3.5-Omni** (Apr 2026) scales to a 256K context and over ten hours of audio input.
- Speech and audio processing get their own treatment later in the course.



Source: Xu et al., "Qwen3-Omni Technical Report", [arXiv:2509.17765](https://arxiv.org/abs/2509.17765), Figure 2.

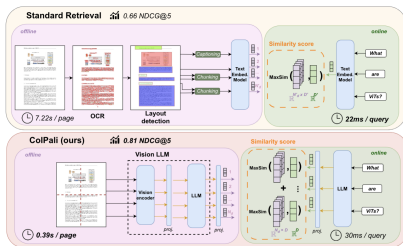
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- Invoices, forms, tables, charts and scanned pages are where most users meet these models. DocVQA and ChartQA (covered earlier) defined the tasks; document parsing is now a headline capability of every flagship.
- **Qwen2.5-VL** turns document pages into structured output and analyzes charts, diagrams and layouts.
- **OmniDocBench** (CVPR 2025) evaluates end-to-end document parsing across nine source types, from academic papers and textbooks to handwriting and densely typeset newspapers.
- Frontier labs report against it: OpenAI cites GPT-5.4 cutting its OmniDocBench error to 0.109 from GPT-5.2's 0.140.

Ouyang et al., [arXiv:2412.07626](https://arxiv.org/abs/2412.07626); OpenAI, "Introducing GPT-5.4", March 2026, openai.com/index/introducing-gpt-5-4.

- Retrieval-augmented generation (covered earlier) assumes clean text. Real corpora arrive as page images whose figures, tables and layout an OCR pipeline flattens or loses.
- **ColPali** (ICLR 2025) embeds the page image directly: a vision-language model produces multi-vector embeddings, matched to the query with ColBERT-style late interaction.
- It outperforms the OCR pipelines while indexing pages an order of magnitude faster: 0.39s against 7.22s per page in the paper's comparison.

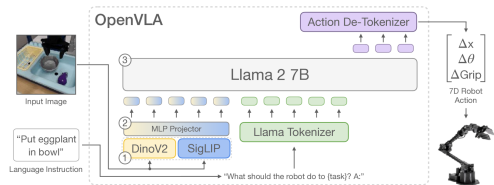


Source: Faysse et al., "ColPali", ICLR 2025, [arXiv:2407.01449](https://arxiv.org/abs/2407.01449), Figure 1.

- **The perception task:** map an instruction to the right pixel. ScreenSpot (ACL 2024) tests grounding on 600+ screenshots across mobile, desktop and web; ScreenSpot-Pro moves to high-resolution professional software, where the best grounding model at its release managed 18.9%.
- **The benchmark:** OSWorld (NeurIPS 2024): 369 real tasks in real operating systems, scored by executing the resulting state.
- **The products:** Claude computer use (Oct 2024, the first frontier model to offer it), OpenAI Operator (Jan 2025) and Gemini 2.5 Computer Use (Oct 2025) all run the same loop: screenshot in, click or keystroke out.
- **The trajectory:** Anthropic reports its models moving from under 15% on OSWorld in late 2024 to 72.5% by early 2026. The agent architectures around this loop are covered later in the course.

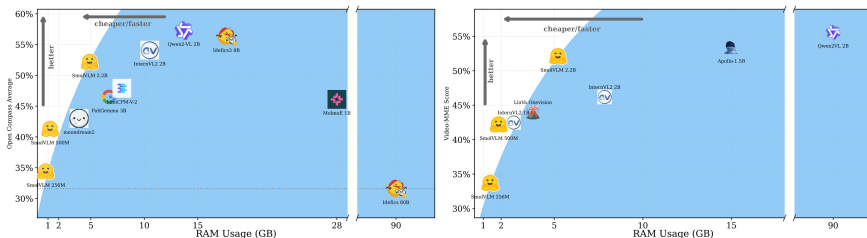
Cheng et al., [arXiv:2401.10935](https://arxiv.org/abs/2401.10935); Li et al., [arXiv:2504.07981](https://arxiv.org/abs/2504.07981); Xie et al., [arXiv:2404.07972](https://arxiv.org/abs/2404.07972); Anthropic, Feb 2026, anthropic.com/news/acquires-vercept.

- **RT-2** (CoRL 2023): fine-tune a VLM on robot trajectories with actions written as text tokens; web-scale knowledge transfers to manipulation.
- **OpenVLA** (CoRL 2024): open 7B model trained on 970k demonstrations; fused DINOv2 and SigLIP features into a Llama 2 backbone, 7-D actions out.
- π_0 (2024): a 300M flow-matching action expert on a PaliGemma VLM emits continuous action chunks; $\pi_{0.5}$ (2025) generalizes to unseen homes.
- **Gemini Robotics** (2025): Gemini 2.0 as a VLA; the 1.5 release pairs an embodied-reasoning VLM with a separate acting VLA it instructs.



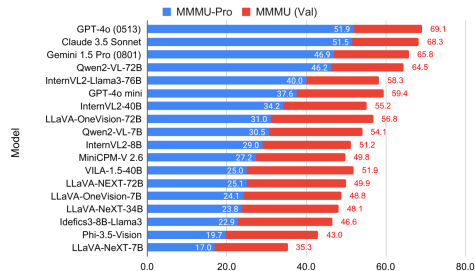
Source: Kim et al., "OpenVLA", CoRL 2024, [arXiv:2406.09246](https://arxiv.org/abs/2406.09246), Figure 1.

- **SmolVLM** (2025): 256M, 500M and 2.2B models; the smallest runs in under 1 GB of GPU memory.
- **PaliGemma 2** (2024) and **Gemma 3** (2025): SigLIP encoders on Gemma LLMs; Gemma 3's 4B, 12B and 27B models share one frozen 417M-parameter encoder.
- **Most visual tokens are wasted anyway:** FastV (ECCV 2024) prunes half of them after layer 2, training-free, cutting FLOPs by 45% with negligible loss.



Benchmark average against GPU memory. Source: Marafioti et al., "SmolVLM", [arXiv:2504.05299](https://arxiv.org/abs/2504.05299), Figure 1.

- **MMMU** (CVPR 2024): 11.5K college-level questions across six disciplines, 30 subjects and 30 image types, from charts and maps to music sheets and chemical structures. GPT-4V managed 56% at launch; the strongest human-expert baseline is 88.6%.
- **MMMU-Pro** (ACL 2025) filters out questions text-only models can answer, widens choices to ten, and adds a vision-only setting where the question itself is an image. Scores drop by 16.8 to 26.9 points.
- By mid-2026 the best systems reach the mid-80s on MMMU and the low-80s on MMMU-Pro, brushing the estimated 85.4% human-expert baseline there.

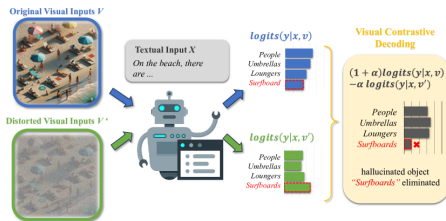


The same models on both benchmarks: every model loses ground on MMMU-Pro. Source: Yue et al., "MMMU-Pro", ACL 2025, [arXiv:2409.02813](https://arxiv.org/abs/2409.02813) (v1), Figure 1.

- **Object hallucination.** Multimodal LLMs describe objects that are not in the image, especially objects that are frequent in, or co-occur with, what they saw in training. POPE (EMNLP 2023) measures this with polling-style yes/no probes.
- **Visual tokens are expensive.** A long webpage screenshot costs 11,427 tokens in Qwen3-VL's own example. DeepSeek-OCR (Oct 2025) inverts the problem, rendering text as an image and reading it back: at a compression ratio below $10\times$ it decodes with 97% precision ("contexts optical compression").
- **Evaluation is fragile.** Benchmark questions leak into training data, and multiple-choice scores overstate perception: MMMU-Pro's vision-only setting exists because models were often reading the question, not the image.
- **The attack surface grows with each modality:** instructions can be embedded in images and audio where text-level filters never look. Safety for agentic systems is covered later in the course.

Li et al., [arXiv:2305.10355](#); Wei et al., [arXiv:2510.18234](#).

- **Fix the data:** RLHF-V (CVPR 2024) collects segment-level human *corrections* of hallucinated spans and trains on them with dense preference optimization; 1.4k fine-grained corrections already cut object hallucination sharply.
- **Fix the decoding:** Visual Contrastive Decoding (CVPR 2024) needs no training at all: contrast the token distribution given the real image against the one given a noised copy, and the language prior's contribution cancels out.



Source: Leng et al., "Visual Contrastive Decoding", CVPR 2024, [arXiv:2311.16922](https://arxiv.org/abs/2311.16922), Figure 1.

- **A single image can break alignment:** Qi et al. (AAAI 2024) optimize one adversarial image that universally jailbreaks an aligned vision-language model, eliciting harmful behavior far beyond the small corpus used to build the image.
- **Injection hides in pixels and audio:** Bagdasaryan et al. (2023) blend an instruction-carrying perturbation into an image or sound clip; when a user innocently asks about the file, the model follows the hidden instruction.
- **Why it works:** safety training lives largely in text space, while the continuous, high-dimensional visual channel routes around it.
- Defenses lag the attacks; a deployed multimodal system has to treat every pixel as untrusted input.

Qi et al., [arXiv:2306.13213](#); Bagdasaryan et al., [arXiv:2307.10490](#).

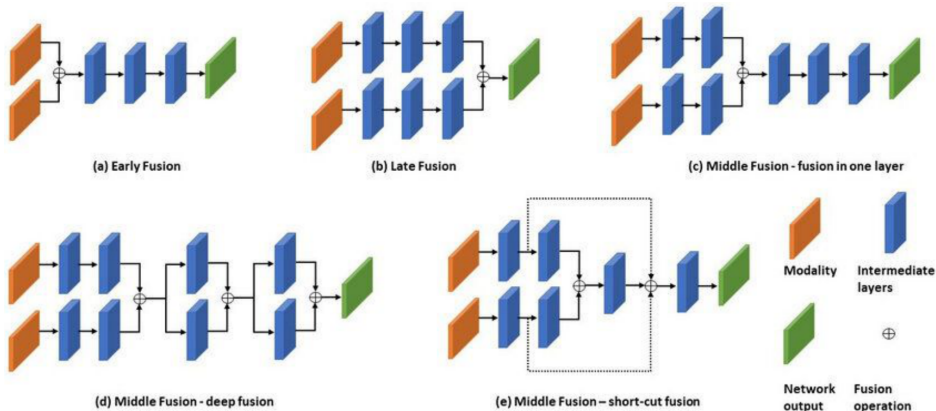
Multimodal Architectures

- ▶ **Early Fusion:** Combine the modalities at the input of a single shared model.
Then: VisualBERT added detector regions as input tokens. Now: Chameleon and Llama 4 pretrain on mixed token sequences from the start.
- ▶ **Late Fusion (dual encoder):** Process each modality separately, then merge only at the output.
CLIP and SigLIP: right for retrieval and zero-shot scoring, too weak for generation.
- ▶ **Middle Fusion / Cross-modal Attention:** Let the modalities interact inside the stack.
Flamingo's gated cross-attention; the projector recipe merges at the token level instead, early fusion around an unchanged LLM.

Design Considerations:

- ▶ **Alignment:** How to align representations across modalities.
- ▶ **Modality Dropout:** Handling missing or noisy modalities during training or inference.
- ▶ **Shared vs. Modality-Specific Layers:** Deciding which layers are shared and which are unique to each modality.

Where Fusion Happens



Orange is a modality, blue an intermediate layer, green the network output, and $\oplus$ a fusion operation. Middle fusion is the general case: the hybrid, cross-modal-attention designs sit here. Source: Feng et al., "Deep Multi-modal Object Detection and Semantic Segmentation for Autonomous Driving", [arXiv:1902.07830](https://arxiv.org/abs/1902.07830), Figure 8.

MultiModal Learning Challenges



Representation

Finding a good representation is crucial for the success of multimodal models that combine different modalities such as image, text, and audio.



Fusion

Fusing information from different modalities to perform a prediction task is the core of multimodal learning and can be challenging due to the diverse nature of multimodal data.



Alignment

Aligning data from different modalities to create modality-invariant representations is necessary to combine information effectively.



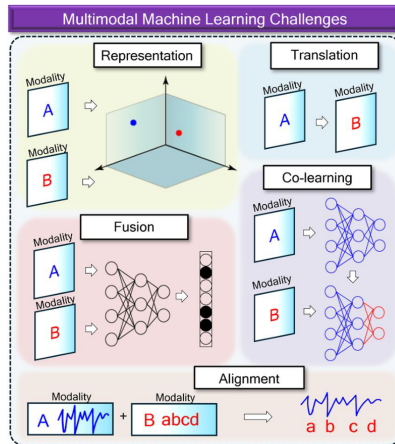
Translation

Translating data from one modality to another is a crucial task in multimodal learning, but can be difficult due to the subjective and open-ended nature of translations.



Co-learning

Co-learning aims to transfer knowledge learned through one or more modalities to tasks involving another. It is useful in medical diagnosis combining CT and MRI scans.



Representation, fusion, alignment, translation and co-learning: the five-way taxonomy of Baltrušaitis, Ahuja & Morency, “Multimodal Machine Learning: A Survey and Taxonomy”, [arXiv:1705.09406](https://arxiv.org/abs/1705.09406). Object hallucination is an alignment failure; omni models are translation at work; the taxonomy predates them all and still fits.

- ▶ **Contrastive alignment** (CLIP, then SigLIP) gives encoders that score and retrieve; conversation took a decoder LLM behind a projector plus visual instruction tuning.
- ▶ **The open recipe scaled** through LLaVA's successors into today's flagships (Qwen3-VL, InternVL3.5): native resolution, grounding, documents, video.
- ▶ **Frontier systems are natively multimodal**: one network reads and writes text, images and audio (GPT-4o onward), reasons by acting on images (o3), and generates images mid-conversation (Nano Banana).
- ▶ **The same perception stack works beyond chat**: retrieving page images, driving GUIs, controlling robots.
- ▶ **Where fusion happens** (early, middle, late) is still the design axis; at scale, early fusion concedes nothing.
- ▶ **Benchmarks brush the human-expert ceiling** (MMMU-Pro in the low-80s); hallucination, token cost, jailbreaks and fragile evaluation stay open.
- ▶ The agentic systems built on these models, their loops, tools and safety questions, are covered later in the course.

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